Fiscal Year:	FY 2023	**Task Last Updated:**	FY 11/05/2022
PI Name:	Plawsky, Joel Sc.D.		
Project Title:	LOOP-based Constrained Vapor Bubbles		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017 Physical Sciences NNH17ZTT001N-17PSI-E: Use of the NASA Physical Sciences Informatics System – Appendix E
Start Date:	11/01/2019	**End Date:**	03/31/2023
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/31/2023 per NSSC information (Ed., 11/3/22) NOTE: End date changed to 10/31/2022 per PI; original end date was 10/31/2021 (Ed., 8/17/21)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
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Task Description:	Loop heat pipes are quite common heat transfer devices in satellites and space systems but often fail and the causes are not always readily apparent. We had built a prototype loop heat pipe as part of the original Constrained Vapor Bubble (CVB) program but only used it to look at refrigerants and especially to help understand meniscus stability. In the proposal, we would like to resurrect that system, based on a wickless, square cross-section CVB and use it to collect ground-based data that we can compare with the linear heat pipe CVB data obtained during CVB1 and CVB2 missions. In particular, we are concerned with mapping the vapor liquid interface, driving the system to the point where Marangoni flows become important, and understanding the mechanisms of failure or instability in wickless loop systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work will enhance the development of heat pipes which are not only used for space applications but also for cooling high heat generation devices on Earth. We are also augmenting and enhancing our interferometry techniques which will greatly expand the types of fluids and surfaces that we can investigate . This will have great utility in looking at the fundamental mechanisms governing phase change heat and mass transfer. Since phase change is a key process in distillation, power generation, refrigeration, and electronics cooling, the more we can learn and uncover the more efficient devices we will be able to manufacture. We have also expanded the scope of the project looking into simulations of evaporation and condensation on surface and molecular dynamics simulations of surface tension of fluid mixtures. The work will help expand our knowledge of how fluids evaporate on surfaces, why menisci oscillate during evaporation, and why mixtures of liquid may make for more efficient heat pipes than pure fluids alone. In addition the simulations may also point to the intermolecular mechanisms by which self rewetting fluids operate.
Task Progress:	We have built and instrumented a loop heat pipe system. This year it will be filled and tested. We have developed our interferometry technique so that we can investigate surfaces of low reflectivity. This provides the tools we need to understand how nm-thick coatings will affect evaporation and condensation processes on surfaces. Much work has been done on alternative surfaces but the experiments are generally on an ad-hoc basis without a fundamental understanding of how intermolecular forces govern these processes or how changing the refractive index and thickness of a nm coating can influence the surface properties and thermal transport behavior. Data have been taken using ethanol as the working fluid and are currently being analyzed. We have also been working on some molecular dynamics calculations of surface tension based on data taken that show the mixture of pentane and isohexane that we use in our experiments exhibits non-ideal behavior. All previous evidence suggested that the surface tension of these mixtures varies linearly with composition. So far, the molecular dynamics simulations are not sensitive enough to pick up the experimental behavior, but that is common. It is difficult to get exact values of the macroscopic property using a molecular simulation. What we are picking up is a change in the orientation of molecules at the interface. In particular, isohexane looks to have a preferred orientation with the portion of the molecule having the highest density of carbon atoms pointing away from the liquid/vapor interface. This preferred orientation may be one way that the mixture composition changes the surface tension. The preferential orientation disappears at high temperature. We have also been simulating evaporation and condensation on surfaces where we oscillate the surface temperature to simulate meniscus oscillations we observe when we heat the liquids. There does appear to be a preferential frequency of oscillation that maximizes the evaporative heat transfer, but that depends upon the liquid used, the type of substrate, and the degree of temperature swing of the surface. If the surface temperature oscillates and cools too much, evaporative heat transfer is not favored; however, if the surface temperature oscillates but does not dip below the saturation temperature, it is possible to still have condensation due to the interface curvature and strength of the intermolecular forces, and that enables higher evaporative heat transfer than having a steady meniscus shape. Finally, during the pandemic when we were locked down, we began thinking about how to represent and optimize wick structures in heat pipes and vapor chambers. This led us to look to mazes and geometrical abstractions for complicated surfaces and to investigate the transport properties of these kinds of geometries. We have shown that maze systems can mimic geometries that have a stochastic component to them and that we can use the elements of graph and network theories to analyze and represent their properties. We have applied this to mimic a commercial membrane and have extended the analysis to thermal systems. We have used our analyses to better understand the effect of geometry on transport. This had led to us developing virtual reality simulations that students can use to understand the process and we can design and print our structures for experimental testing using additive manufacturing techniques. The latter opens the way toward developing new laboratory modules that students can use to explore transport and design wicks, separations equipment, thermal interface materials, and many other distributed types of transport systems.
Bibliography Type:	Description: (Last Updated: 10/14/2025)
Articles in Peer-reviewed Journals	Yu J, Pawar A, Plawsky JL, Chao DF. "The effect of bubble nucleation on the performance of a wickless heat pipe in microgravity." npjMicrogravity. 2022 Apr 28;8(1):1-10. https://doi.org/10.1038/s41526-022-00197-5 , Apr-2022
Articles in Peer-reviewed Journals	Guo A, Marshall WC, Woodcock CC, Plawsky, J.L. "Transport in mazes; simple geometric representations to guide the design of engineered systems." Chemical Engineering Science. 2022 Mar 15;250:117416. https://doi.org/10.1016/j.ces.2021.117416 , Mar-2022
Articles in Peer-reviewed Journals	Wayner PC, Plawsky JL. "Revew: Change-of-phase in an extended meniscus 2020 Max Jakob Memorial Award Paper." J. Heat Transfer. Jan 2022;144(1):013101. https://doi.org/10.1115/1.4052688 , Jan-2022